



Certified Penetration Testing Professional

Metasploit Cheat Sheet

Metasploit

Source: <https://www.metasploit.com>

Metasploit is an open-source project that provides the infrastructure, content, and tools to perform penetration tests and extensive security auditing. It is a tool that provides information about security vulnerabilities and aids in penetration testing. Metasploit framework is also used for developing and executing exploits which promotes in gaining remote access to a system by exploiting any vulnerability present in that server. Meterpreter is a payload inside the framework.

The following table lists the various Metasploit commands and their respective scanning methods.

1. [Metasploit General Information](#)
2. [Executing an Exploit / Scanner / Module](#)
3. [Session Handling](#)
4. [Using Database](#)
 - i. [First Time Setup \(Linux command line\)](#)
 - ii. [Inside msfconsole](#)
5. [Meterpreter Session Commands](#)
 - i. [Base Commands](#)
 - ii. [Process Commands](#)
 - iii. [File System Commands](#)
 - iv. [Escalate Privileges](#)
 - v. [Networking Commands](#)
 - vi. [Additional Commands](#)

6. [Session Management](#)
7. [Interface / Output commands](#)
8. [Msfvenom Command Options](#)
9. [Important Auxiliary Modules](#)

1. Metasploit General Information

Metasploit Command	Description
<code>msfconsole</code>	Launch program
<code>version</code>	Display current version
<code>msfupdate</code>	Pull weekly update
<code>makerc <FILE.rc></code>	Saves recent commands to a file
<code>msfconsole -r <FILE.rc></code>	Loads resource file to automate Metasploit commands

2. Executing an Exploit / Scanner / Module

Metasploit Command	Description
<code>use <MODULE></code>	Set the exploit to use
<code>set payload <PAYLOAD></code>	Set the payload
<code>show options</code>	Show all options
<code>set <OPTION> <SETTING></code>	Set a specific option setting
<code>exploit or run</code>	Execute exploit

3. Session Handling

Metasploit Command	Description
<code>sessions -l</code>	List all sessions
<code>sessions -i <ID></code>	Interact with a session
<code>background</code> or <code>^Z</code>	Detach from session

4. Using Database

i. First Time Setup (Linux command line)

Metasploit Command	Description
<code>service postgresql start</code>	Start the PostgreSQL service
<code>msfdb init</code>	Initialize the Metasploit database

ii. Inside msfconsole

Metasploit Command	Description
<code>db_status</code>	Display connected status
<code>hosts</code>	Display hosts in database
<code>services</code>	Show ports in database
<code>vulns</code>	Exhibit all vulnerabilities

5. Meterpreter Session Commands

i. Base commands

Metasploit Command	Description
<code>sysinfo</code>	Display system name and OS type
<code>shutdown / reboot</code>	Shutdown or reboot the system

<code>exit / quit</code>	Exit Meterpreter session
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ii. Process Commands

Metasploit Command	Description
<code>ps</code>	Show running processes list
<code>kill <PID></code>	Terminate process
<code>getuid</code>	Show user ID
<code>getpid</code>	Show process ID that Meterpreter is running inside
<code>migrate <PID></code>	Migrate to another process
<code>execute</code>	Execute given program with the privileges of the process

iii. File System Commands

Metasploit Command	Description
<code>pwd / lpwd/getwd</code>	Display current working directory (local / remote)
<code>cd</code>	Change directory
<code>lcd</code>	Change directory (local)
<code>mkdir</code>	Make directory
<code>rmdir</code>	Remove directory
<code>cat</code>	Show contents of a file
<code>edit <FILE></code>	Edit a file in default editor (vi)
<code>upload / download</code>	Upload / download a file from /to target machine

iv. Escalate Privileges

Metasploit Command	Description
<code>use priv</code>	Load the privilege escalation extension
<code>getsystem</code>	Gain administrative-level privileges
<code>getprivs</code>	Elevate privileges

v. Networking Commands

Metasploit Command	Description
<code>ipconfig</code>	Show network interface information
<code>route</code>	Manage/view the system's routing table
<code>C</code>	Forward packets through TCP session
<code>route add <Target IP/ Subnet></code>	Pivot through session by adding route in MSF
<code>route remove <Target IP/ Subnet></code>	Delete route inside MSF
<code>sniffer</code>	Allow network sniffing interaction commands
<code>portfwd</code>	Port forwarding connections
<code>portfwd -L <Local host></code>	Local host to listen
<code>portfwd -l <Local port></code>	Local port to listen
<code>Portfwd -p <Remote port></code>	Remote port to connect
<code>portfwd -r <Remote host></code>	Remote host to connect

vi. Additional Commands

Metasploit Command	Description
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<code>shell</code>	Drop into a shell on the target machine
<code>hashdump</code>	Show all password hashes in Windows
<code>idletime</code>	Display idle time of the machine
<code>screenshot</code>	Save the screenshot
<code>clearev</code>	Clear the logs
<code>uictl [enable/disable] [keyboard/mouse]</code>	Enable or disable the mouse or keyboard of the machine
<code>use</code>	Extension load
<code>channel</code>	Display active channel
<code>reg</code>	Access machine registry
<code>steal_token</code>	Attempts to steal impersonation token from target
<code>espia</code>	Desktop spying by screenshots
<code>incognito</code>	Impersonation commands
<code>msf> search</code>	Search for any module
<code>msf > use exploit</code>	Specify and exploit to use

6. Session Management

Metasploit Command	Description
<code>msf > exploit -z</code>	Run exploit in background expecting one session
<code>msf > sessions -i [SessionID]</code>	Interact with backgrounded session
<code>msf > exploit -j</code>	Run exploit in background expecting one or more sessions
<code>msf > sessions -l</code>	List all backgrounded sessions

<code>msf > jobs -l</code>	List all current jobs
<code>msf > jobs -k [JobID]</code>	Kills job
<code>meterpreter > <Ctrl+Z> / meterpreter > background</code>	Background current interactive session

7. Interface / Output Commands

Metasploit Command	Description
<code>enumdesktops</code>	Display all existing desktops
<code>getdesktop</code>	Display current desktop
<code>keyscan_start</code>	Start keylogger in target machine
<code>keyscan_stop</code>	Stop keylogger in target machine
<code>set_desktop</code>	Configure desktop
<code>keyscan_dump</code>	Dump keylogger content

8. Msfvenom Command Options

Metasploit Command	Description
<code>-p (Payload option)</code>	Show payload standard options
<code>-l (list type)</code>	List module type
<code>-f (format)</code>	Output format
<code>-e (encoder)</code>	Define which encoder to use
<code>-a (Architecture or platform)</code>	Define which platform to use
<code>-s (Space)</code>	Define maximum payload capacity
<code>-b (characters)</code>	Define set of characters not to use

-i (Number of times)	Define number of times to use encoder
-x (File name)	Define a custom file to use as template
-o (output)	Save payload
-h	Help

9. Important Auxiliary Modules

Metasploit Command	Description
<code>msf > use auxiliary/scanner/portscan/tcp</code> <code>msf > set RHOSTS <Target IP/Subnet></code> <code>msf > set PORTS 1-1000</code> <code>msf > run</code>	Port scanning module
<code>msf > use auxiliary/gather/dns_enum</code> <code>msf > set DOMAIN target.tgt</code> <code>msf > run</code>	DNS Enumeration module
<code>msf > use auxiliary/server/ftp</code> <code>msf > set FTPROOT /tmp/ftproot</code> <code>msf > run</code>	FTP Server module
<code>msf > use auxiliary/server/socks4</code> <code>msf > run</code>	Proxy Server module
<code>msf > use auxiliary/scanner/snmp/snmp_enum</code> <code>msf > set RHOSTS <Target IP></code> <code>msf > exploit</code>	SNMP Enumeration module
<code>msf > use auxiliary/scanner/sip/enumerator</code> <code>msf > set RHOSTS <Target IP/Subnet></code> <code>msf > run</code>	SIP Enumeration module
<code>msf > use auxiliary/scanner/ftp/ftp_version</code> <code>msf > set RHOSTS <Target IP></code> <code>msf > exploit</code>	FTP Enumeration module
<code>msf > use</code> <code>auxiliary/scanner/discovery/arp_sweep</code> <code>msf > set RHOSTS <Target IP-Range></code> <code>msf > set SHOSTS <Target IP></code> <code>msf > set SMAC <MAC Address></code>	ARP Sweep module

<pre>msf > set THREADS <Number of concurrent threads> msf > run</pre>	
<pre>msf > use auxiliary/scanner/discovery/ipv6_neighbor msf > set RHOSTS <Target IP-Range> msf > set SHOSTS <Target IP> msf > set SMAC <MAC Address> msf > set THREADS <Number of concurrent threads> msf > run</pre>	IPV6 Neighbor module
<pre>msf > use auxiliary/scanner/discovery/udp_probe msf > set RHOSTS <Target IP-Range> msf > set THREADS <Number of concurrent threads> msf > run</pre>	UDP Probe module
<pre>msf > use auxiliary/scanner/discovery/udp_sweep msf > set RHOSTS <Target IP-Range> msf > set THREADS <Number of concurrent threads> msf > run</pre>	UDP Sweep module
<pre>msf > use auxiliary/scanner/scada/modbus_findunitid msf > set RHOSTS <Target IP> msf > run</pre>	Scan and detect Modbus Slaves
<pre>msf > use x86/opty2 msf nop(opty2) > generate -h Usage: generate [options] length</pre>	Generates a NOP sled of a given length